SDG Blockchain Accelerator Cohort 2
Deliverables Roadmap

Accelerator Duration: 29 September - 15 December 2025
Challenge Owner (CO): CCIT-Office of Procurement
Solution Maker (SM): Vaka

1. Challenge Definition

1.1 Refined Challenge Statement

Many African countries face challenges in maintaining hydro-meteorological stations and sharing data, often trapped in an "install-and-decay" cycle where infrastructure fails due to lack of sustainable maintenance funding.

This limits global forecasting accuracy and disrupts the global objective to enhance community resilience with strengthened Early Warning Systems worldwide. This is especially a problem in Least Developed Countries and Small Island Developing States, where the "missing middle" is a transparent verification mechanism to link funding directly to data quality.

1.2 Local Context

For example, in Tanzania, while the country is implementing a **$13.9 million SOFF Investment Phase** to upgrade its network, limited resources still hinder the long-term maintenance of this infrastructure. The **Tanzania Meteorological Authority (TMA)** faces high operational costs for its surface and upper-air stations, often struggling to meet **Global Basic Observing Network (GBON)** standards consistently. Furthermore, there is a structural need to balance the TMA's statutory mandate for "commercial services" (TMA Act 2019) with the global push for open data sharing.

1.3 Alignment with UNDP Country Office and National Development Priorities

Explain how the proposed solution aligns with the strategic objectives of the UNDP Country Office and national development frameworks. Does it support existing programs or government priorities (e.g., digital transformation, climate resilience, youth employment, financial inclusion)? Highlight any synergies with UNDP initiatives, government strategies, or donor-funded efforts.

The proposed blockchain solution is highly aligned with the **UNDP's Digital Strategy 2022-2025**, specifically the focus on building inclusive "Digital Public Infrastructure" that fosters trust. It directly supports national development priorities for climate resilience (Third National Five-Year Development Plan) by strengthening the foundational weather data layer necessary for effective Early Warning Systems. Crucially, the initiative exhibits strong synergy with the **Systematic Observations Financing Facility (SOFF),** a UN Multi-Partner Trust Fund (UNMPTF) co-founded with UNDP/WMO/UNEP partnership. It acts as a "digital operating system" for the SOFF Compliance Phase, creating a layer of trust and traceability for **results-based payments**. This ensures the long-term sustainability of the TMA's network by proving, via a "Shadow Payment" pilot, that data quality can be verified and monetized without immediate regulatory risk.

This ensures the long-term sustainability and compliance of a nation's weather observation network with the Global Basic Observing Network (GBON).

1.4 Relevance to UNDP Programmes and Resource Mapping

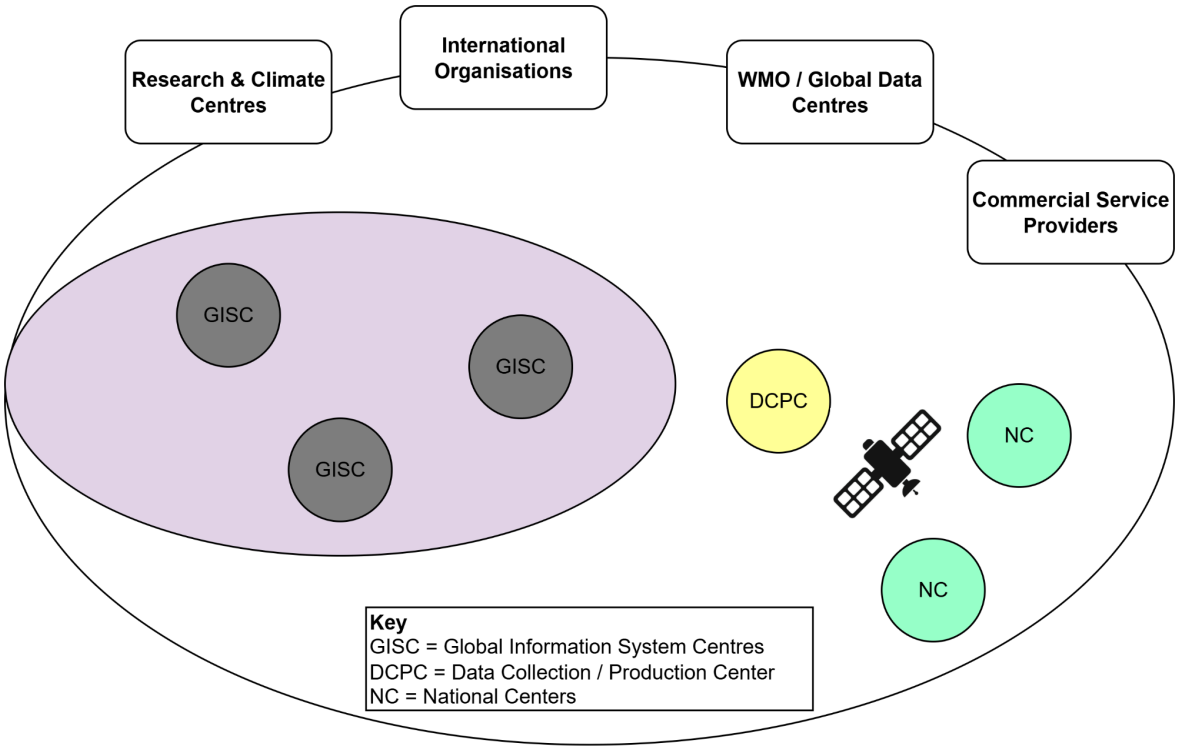
(Is the pilot building on an existing initiative, or is it a new standalone project? If the project builds on an existing initiative, provide further details. Include scope, scale, timeline and available co-financing or in-kind contributions.

This initiative aligns with UNDP's goals of climate resilience, digital innovation, and inclusive development. It builds directly on **SOFF** and UNDP's support for the **Early Warning for All** programme, sustainable meteorological systems, and data governance.

Specifically, this pilot functions as the **"Digital Compliance Layer"** for the **$13.9 million SOFF Investment Phase** currently underway in Tanzania. While the Investment Phase upgrades physical hardware (stations), this pilot deploys the software infrastructure required to verify that these stations remain operational, directly addressing the "missing middle" in results-based financing.

Scope & Scale:

- **Timeline:** 6-Month "Shadow Payment" Pilot (Non-financial simulation).
- **Target:** Validation of data flow from the 27 Surface Stations and 5 Upper-Air Stations identified in the Tanzania Country Hydromet Diagnostics.
- **Co-Financing & Contributions:**
    - **In-Kind:** TMA provides access to station data streams and engineering staff time for calibration.
    - **Technology:** Vaka Consulting and ClimaFI contribute the core Verification Engine and Blockchain Testnet infrastructure.
    - **Resource Mapping:** The pilot leverages the existing hardware procurement of the SOFF Investment Phase, requiring no new sensor purchases, only digital integration.



1.5 Key Stakeholders & Potential Partners

(Please list all the partners involved in this project. List all relevant parties who will interact with, benefit from, or influence the solution (this may include government agencies, NGOs, community members, or tech partners.)

Donors & Funders (Pay for Results):

- **Systematic Observations Financing Facility (SOFF):** The primary partner for future scaling. The pilot builds the "Automated Compliance Tool" SOFF needs to operationalize its Compliance Phase efficiently across multiple countries
- **Climate Finance Institutions & Bilateral Donors:** Entities holding resources that require reliable, sustained data streams to fund climate resilience initiatives (e.g., adaptation funds)
- **Other Bilateral and Multilateral Donors:** Partners supporting global numerical modelling capacities who need assurance that their "capacity building" grants are delivering lasting data results

Government Agencies:

- **National Meteorological and Hydrological Services (NMHS):** The primary data providers who generate and share observations. They are the direct beneficiaries receiving payments for verified data
- **Ministries of Finance & Environment:** Responsible for approving data-sharing licenses, integrating payments into public budgets, and overseeing national development priorities.The pilot demonstrates how weather data can become a "bankable asset" to supplement national budgets
- **National Meteorological Ministries:** The overarching government bodies managing the meteorological infrastructure

International Organizations (Standards & Convening)

- **World Meteorological Organization (WMO):** Sets the global standards (GBON/WIS2) for data exchange and quality monitoring
- **United Nations Development Programme (UNDP):**
  - **UNDP Tanzania Country Office:** The UNDP Tanzania Country Office (CO) acts as the bridge between international technological innovation and national institutional reform. Under the leadership of the Resident Representative, the CO's mandate is to deliver on the *UN Sustainable Development Cooperation Framework (UNSDCF)* and the *UNDP Digital Strategy 2022-2025*. By hosting this pilot, the CO aims to transition from traditional input-based aid to high-assurance, results-based financing that can be scaled across the Global South.
  - **UNDP Country Offices:** Align the project with national development priorities and coordinate local implementation
  - **UNDP Office of Procurement:** Provides specialized guidance to ensure the solution creates a transparent, best-value procurement model for future data purchases

Technology Partners (Verification & Payout Rails)

- **ClimaFI:** Provides the "Verification Oracle." Responsible for the off-chain verification engine that checks data against WMO standards before it touches the blockchain
- **Vaka Consulting:** The blockchain execution partner. Responsible for the "Shadow Ledger," smart contracts, and eventually the live tokenized finance rails
- **Cardano:** The Layer 1 blockchain infrastructure chosen for its high assurance, determinism, and low transaction costs
- **EMURGO:** Strategic blockchain technology provider supporting the ecosystem's technical scalability

Local & Beneficiary Stakeholders

- **Rural Communities:** Vulnerable populations who are the ultimate beneficiaries of the improved Early Warning Systems driven by more reliable data
- **Regional Climate Centers & Universities:** Institutions that benefit from improved data availability and capacity building
- **Commercial Service Providers:** Future buyers (e.g., agricultural insurers like Pula) who can use the "Verified Data Tokens" for index-based insurance products, creating a commercial sustainability path for TMA

1.6 Expected Impact

(Outline intended outcomes from the CO's perspective. What would a successful pilot enable - policy change, improved services, empowerment, systems improvement, or scaling innovation?)

Pilot Phase Outcomes (Exploratory & De-risking: 6 Months)

- **Validated Data Verification Framework:** Successful deployment of the "Data Ingestion" and "Verification Gate" mechanisms to test the feasibility of automatically checking TMA station data against GBON/WIS2 standards in a real-world environment

- **Proof of Value (Shadow Accounting):** Demonstration of a "shadow payment" system that calculates and visualizes the potential earnings for the TMA based on data quality, providing a risk-free environment to explore how results-based incentives influence operational behavior
- **Immutable Performance Audit:** Generation of "Data Receipt Tokens" for valid datasets, creating a transparent, tamper-proof record of historical performance that can be used to build trust with future international donors
- **Institutional Readiness & Learning:** Capacity building for TMA and UNDP staff through hands-on use of dashboards, ensuring that the technology is "institutionally absorbable" before any transition to live financial incentives

Long-Term & Systemic Impact (Strategic Vision)

- **Operationalizing Results-Based Finance (RBF):** Transitioning from shadow accounting to live settlements, providing a predictable, automated revenue stream for NMHS to fund critical station maintenance and operations
- **Sustainable Infrastructure Lifecycle:** Moving away from sporadic, input-based project funding to a sustainable model where high-quality data generation directly funds its own infrastructure maintenance, breaking the "install-and-decay" cycle.
- **South-South Scaling:** Replicating the validated technical stack from Tanzania to other LDCs and SIDS to close the global data gap, ultimately enhancing the accuracy of global climate models and Early Warning Systems.
- **Global Policy Transformation:** Shifting the global standard for climate finance toward transparent, high-assurance "Pay-for-Results" models, where funding is algorithmically triggered by verified impact rather than upfront grants.

At the end of the pilot, we expect to have demonstrated how to:

- Improve the reliability and coverage of national meteorological data
- Create a blueprint for sustainable financing of national hydrological and meteorological services
- Empower local institutions with advanced data governance tools and technical capacity

1.7 Target SDGs and Indicators

(List up to three specific SDG indicators your challenge and proposed solution will directly contribute to. Focus on indicators you will actively address, not indirect ones.)

This initiative contributes to:

Goal	Target	Specific Indicator
SDG 13: Climate Action	**13.1** Strengthen resilience and adaptive capacity	**13.1.2:** Number of countries that adopt and implement national disaster risk reduction strategies

SDG 9: Industry, Innovation & Infrastructure	**9.1** Develop quality, reliable, sustainable and resilient infrastructure	**9.1.1:** Proportion of the rural population who live within 2 km of an all-season road (Adapting this to **weather station density** for climate-resilient infrastructure)
SDG 17: Partnerships for the Goals	**17.6** Enhance North-South and South-South cooperation	**17.6.1:** Number of science and/or technology cooperation agreements and programmes between countries, by type of cooperation

2. User Mapping & Testing Strategy

2.1 Primary User Persona

Attribute	Details
Name	**SOFF** (Systematic Observations Financing Facility) and **National Hydrological and Meteorological Services (NHMS)** for SOFF compliance Phase.
Role (NHMS)	**Core Role:** Capturing metrics of quality-controlled data from sensors across the country and sharing it with the WMO and global producing centers via GTS/WIS2. **The Business Case for the Pilot:** The solution provides an automated accounting system to measure all data shared. This allows for faster and more seamless processing of payments from SOFF and other donors by providing "Proof of Work" for the climate sector. **Operational Environment:** Government offices often manage aging weather stations with limited digital infrastructure and constrained budgets.

Role (SOFF)	Providing verifiable result-based payments to support NHMS for sustained Global Basic Observing Network (GBON) compliance. In other words, the success of countries and SOFF is measured by the amount and quality of internationally exchanged data.
Environment	NHMS operates out of a government office with limited digital infrastructure, overseeing a network of aging weather stations across the country.

Strategic Goals for the NMHS:

- **Data Excellence:** Promoting consistent, high-quality collection that meets international GBON standards
- **Financial Sustainability:** Establishing a transparent methodology with donors to release automatic payments for station maintenance and staff training
- **Trust Foundations:** Using the "Shadow Payment" model to agree on an accounting system before moving to live financial transactions

Primary Challenges:

- **Budget Gaps:** Insufficient funds for daily operations, equipment upgrades, and the high cost of data transmission
- **Compliance Burden:** Difficulty in meeting rigorous international data frequency standards without automated reporting tools
- **Climate Finance Access:** Limited existing pathways to access results-based climate finance mechanisms

2.2 User Story

As a National Meteorological Director,

I want to receive immediate, verifiable digital receipts and a calculated financial valuation whenever my ministry shares weather data with global centers,

So that we can provide mathematical proof of the economic value of our data compliance to international donors, validate our operational performance, and build the evidence base for future automated infrastructure funding.

Success Statement: Timely verification and simulated payments support our data-sharing continuity. By proving our compliance hourly, we can justify the resources needed to fund the maintenance of our sensor networks, ensuring they never fall into the "install-and-decay" cycle.

The Final Outcome: We ensure the long-term sustainability of our national observation infrastructure, meeting our GBON mandates and contributing significantly to global climate resilience.

2.3 Key Stakeholders/Partners

Donors & Funders (Pay for Results):

- **Systematic Observations Financing Facility (SOFF):** A primary potential partner for scaling, providing requirements for **GBON compliance** and reviewing pilot outcomes for future funding potential
- **Climate Finance Institutions & Bilateral Donors:** Entities holding resources that require reliable, sustained data streams to fund climate resilience initiatives
- **Other Bilateral and Multilateral Donors:** Partners interested in supporting global numerical modelling capacities and early warning systems.

Government Agencies

- **National Meteorological and Hydrological Services (NMHS):** The primary data providers who generate and share observations. They are the direct beneficiaries receiving payments for verified data. E.g. Tanzania Meteorological Authority
- **Ministries of Finance & Environment:** Responsible for approving data-sharing licenses, integrating payments into public budgets, and overseeing national development priorities.
- **National Meteorological Ministries:** The overarching government bodies managing the meteorological infrastructure

International Organizations (Standards & Convening)

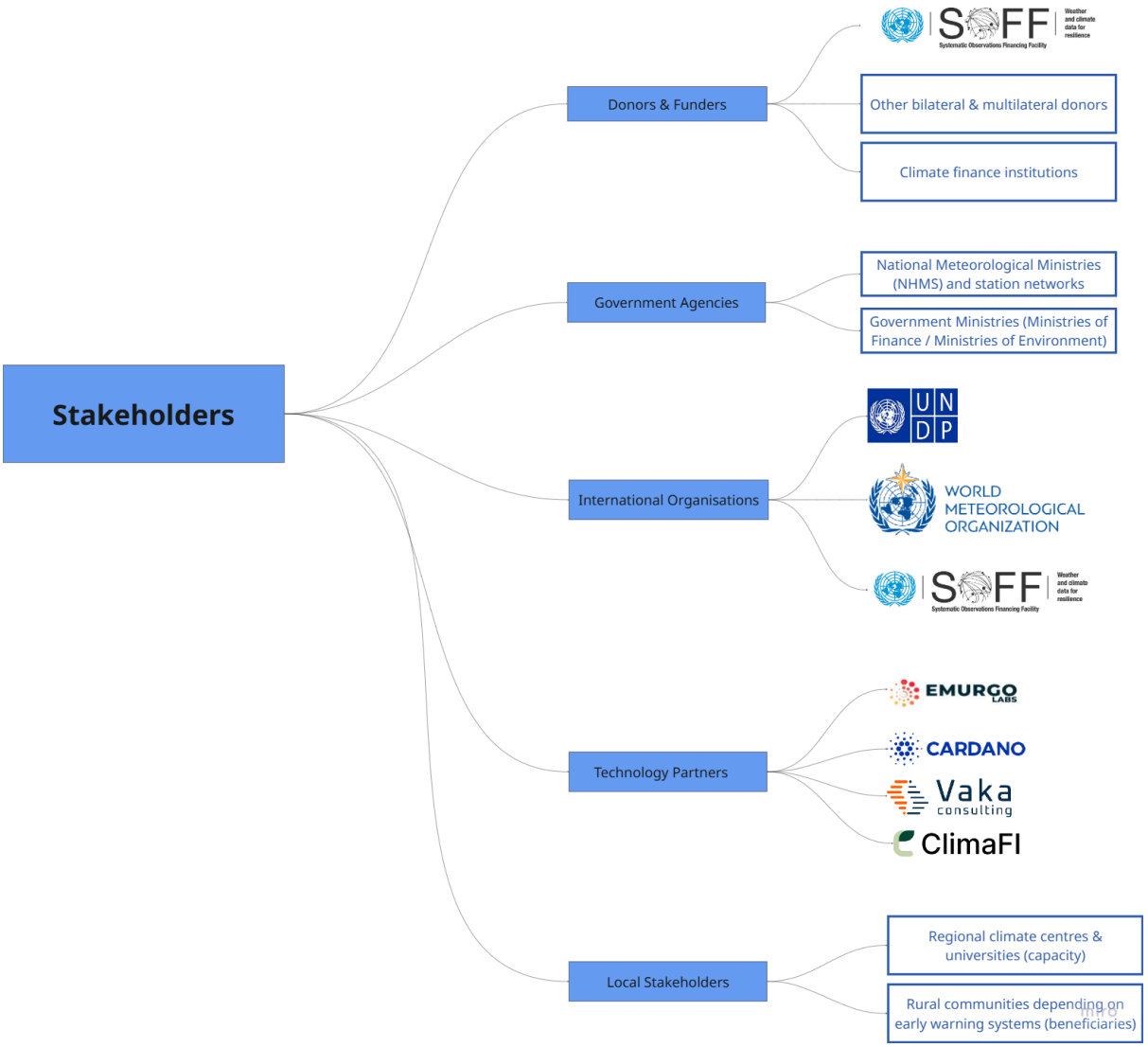
- **World Meteorological Organization (WMO):** Sets the global standards (GBON/WIS2) for data exchange and quality monitoring.
- **United Nations Development Programme (UNDP):**
  - **UNDP Country Offices:** Align the project with national development priorities and coordinate local implementation.
  - **UNDP Office of Procurement:** Provides specialized procurement services for weather monitoring solutions.
  - **The Tanzania Country Office** coordinates local government alignment and facilitates access to ministries.

Technology Partners (Verification & Payout Rails)

- **ClimaFI:** Solution Maker responsible for the off-chain verification engine, anomaly detection, and stakeholder dashboards.
- **Vaka Consulting:** The blockchain execution partner responsible for smart contracts, tokenised finance rails, and Web3 authentication.
- **Cardano:** The Layer 1 blockchain infrastructure provider used for its high assurance and decentralization.
- **EMURGO:** A blockchain technology provider supporting the ecosystem.

Local & Beneficiary Stakeholders

- **Rural Communities:** Vulnerable populations who rely on improved early warning systems for climate resilience.
- **Regional Climate Centers & Universities:** Institutions that benefit from improved data availability and capacity building.
- **Commercial Service Providers:** Potential users or intermediaries in the global weather data market.



2.3 Testing Strategy

How and when will user testing be conducted?

Phase 1: Architecture & Criteria Validation (Weeks 1–2)

- ***Objective:*** *Ensure alignment on the "Rules of the Game" before code deployment.*
- ***Testing Activity:*** *Validation of the* ***Success Criteria Document*** *with UNDP and TMA. Confirm that the defined "Verification Triggers" (e.g., latency <15 mins, valid GBON variables) accurately reflect operational realities.*
- ***Output:*** *Signed-off Architecture and Data Flow protocols.*

Phase 2: Pipeline & Verification Testing (Weeks 3–5)

- ***Objective:*** *Validate that the system can ingest real station data and generate reproducible proofs.*
- ***Testing Activity:***
    - ***Ingestion Stress Test:*** *Feed real historical data from TMA stations into the pipeline to test stability.*
    - ***Hash Verification:*** *Confirm that the system generates identical cryptographic hashes for identical datasets (integrity check).*
    - ***Anomaly Detection:*** *Run "bad data" scenarios to ensure the Verification Gate correctly rejects non-compliant packets.*

Phase 3: Tokenisation & Registry Validation (Weeks 6–7)

- ***Objective:*** *Prove the immutable link between off-chain data and on-chain tokens.*
- ***Testing Activity:***
    - ***Minting Test:*** *Deploy the receipt token policy on the* ***Cardano Testnet****. Verify that valid data triggers the minting of a "Data Receipt Token."*
    - ***Metadata Audit:*** *Check that the token metadata contains the correct* ***Station ID****,* ***Timestamp****, and* ***Data Hash****.*
    - ***Traceability Test:*** *Query the minimal token registry to ensure a specific token can be traced back to its source dataset.*

Phase 4: Dashboard & Simulation Testing (Weeks 8–9)

- ***Objective:*** *Validate the user experience for the Provider (TMA) and the Oversight Body (SOFF).*
- ***Testing Activity:***
    - ***Role-Based Access Control (RBAC):*** *Ensure TMA sees "My Stations" while SOFF sees the "Global Compliance View."*
    - ***Payment Logic Simulation:*** *Test the "Shadow Escrow" engine. Does the system correctly calculate the theoretical payout based on the volume of verified tokens?*
    - ***Aggregated Visibility:*** *Verify that the SOFF dashboard correctly aggregates data across multiple stations to show a network-wide compliance %.*

Phase 5: User Acceptance & Evaluation (Weeks 10–12)

- ***Objective:*** *Assessing readiness for scale and live payments.*
- ***Testing Activity:***
    - ***Stakeholder Walkthrough:*** *Conduct a live demo with TMA and UNDP staff using the "End-to-End" workflow.*
    - ***Feedback Integration:*** *Gather structured feedback on the utility of the* ***SOFF Compliance Dashboard****.*
    - ***Go/No-Go Evaluation:*** *Assess the system against the Phase 1 Success Criteria to determine readiness for Phase 2 (Live Payments).*

Strategic Approach: The "Shadow Payment" Pilot *While our vision is a fully automated results-based finance (RBF) ecosystem, we recognize that shifting*

*institutional fund flows requires trust before automation. Therefore, this PoC is explicitly **non-financial and non-production**.*

*The system functions as a **Digital Twin** of the financial process. It ingests live data, verifies compliance, and mints "Data Receipt Tokens" that trigger simulated payments. This allows SOFF and UNDP to verify the accuracy, speed, and fairness of the system without regulatory risk. Once this "Digital Audit" is proven to match reality, the rails can be upgraded to "Live Settlement" in future phases.*

2.4 Feedback Loop Design

How will feedback from users and stakeholders be gathered and integrated?

Weekly Coordination: Utilize the scheduled weekly meetings between Solution Makers (SMs) and Country Offices (COs) to align on scope and address immediate technical blockers.

Live Feedback Sessions: Dedicate specific sessions following the Week 10 "Stakeholder Demo" to capture structured feedback on UI/UX and reporting workflows directly from NMHS and UNDP staff.

Iterative Refinement: Allocate Week 11 explicitly to "Iteration & Refinement," where developer resources are assigned to fix bugs and implement high-priority changes identified during UAT.

Documentation: Maintain a "Testing & Feedback Documentation" log to track all user inputs and the corresponding "Feedback Integration Summary" to record how the solution evolved.

3. Solution Overview

3.1 Solution Summary

The **Results-Based Payments for Meteorological Data (RBP-MetData)** project builds a "digital trust machine" to bridge the gap between weather data production and climate finance. The solution is defined by four core capabilities:

- **Data Tokenisation:** Introduces a mechanism where meteorological data is automatically verified against GBON/WIS2 standards, with valid data minting a **"Data Receipt Token"** that embeds an immutable cryptographic proof
- **Automated Verification Gate:** Utilizes a rules-based engine (checking determinism, consistency, and timeliness) to ensure only high-quality data generates a token, filtering out anomalies before they reach the payment layer
- **Shadow Payment Logic:** Deploys a smart contract escrow system (operating in "Shadow Mode" for the pilot) that calculates and displays potential payouts based on the volume of valid receipt tokens. This demonstrates the "pay-for-results" model without requiring immediate regulatory clearance for crypto-assets
- **Transparency & Trust:** Establishes a shared, tamper-proof ledger on the Cardano blockchain, reducing audit costs for donors (like SOFF) and providing financial predictability for National Meteorological Services

Operational Workflow The proposed solution ensures that meteorological data provided by national services is both verifiable and financially rewarded.

1. **Ingestion & QA:** Data transmitted from meteorological equipment undergoes a quality assurance process, combining automated standards-based verification (GBON/WIS2 metrics) with optional human oversight
2. **Token Issuance:** Once data passes the "Verification Gate," the system issues a “Data Receipt Token” directly to the wallet of the contributing organisation
3. **Proof of Origin:** Each receipt token embeds a cryptographic proof (hash) of the verified data, creating an immutable link between the token and the underlying dataset. This makes future audits transparent and cost-efficient
4. **Shadow Settlement:** For the pilot, the platform utilizes a "Shadow Escrow" smart contract. Instead of moving live stablecoins, it calculates the *theoretical* revenue earned. This allows Meteorological Organisations to "claim" verified results on a dashboard, proving the concept of Results-Based Financing where payment occurs only when reliable, usable data has been shared.

Future State Vision: In the post-pilot phase, funding organisations (e.g., SOFF) will deposit actual stablecoins into the escrow, and the "Shadow Claims" will become live, automated settlements.

Why This Works: This model reduces verification costs, aligns incentives between funders and providers, and creates a replicable digital infrastructure. By leveraging the blockchain for both data verification and payments, transparency is established, minimizing disputes. As the most high-assurance and decentralized blockchain in operation, **Cardano** provides the critical infrastructure to scale this solution so that high-value climate investments are not subject to platform risk

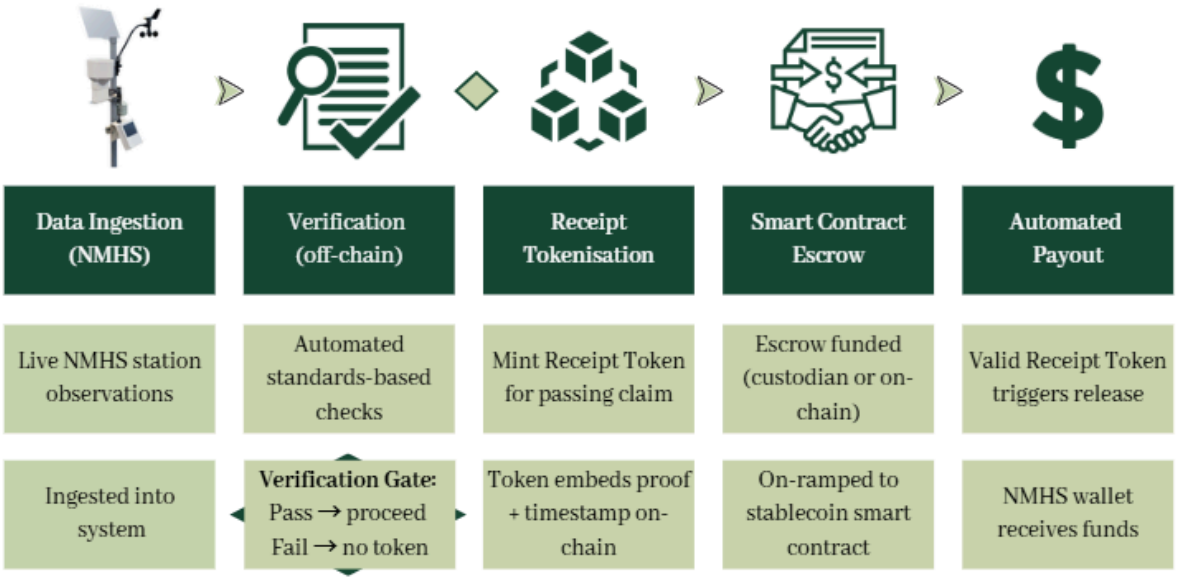
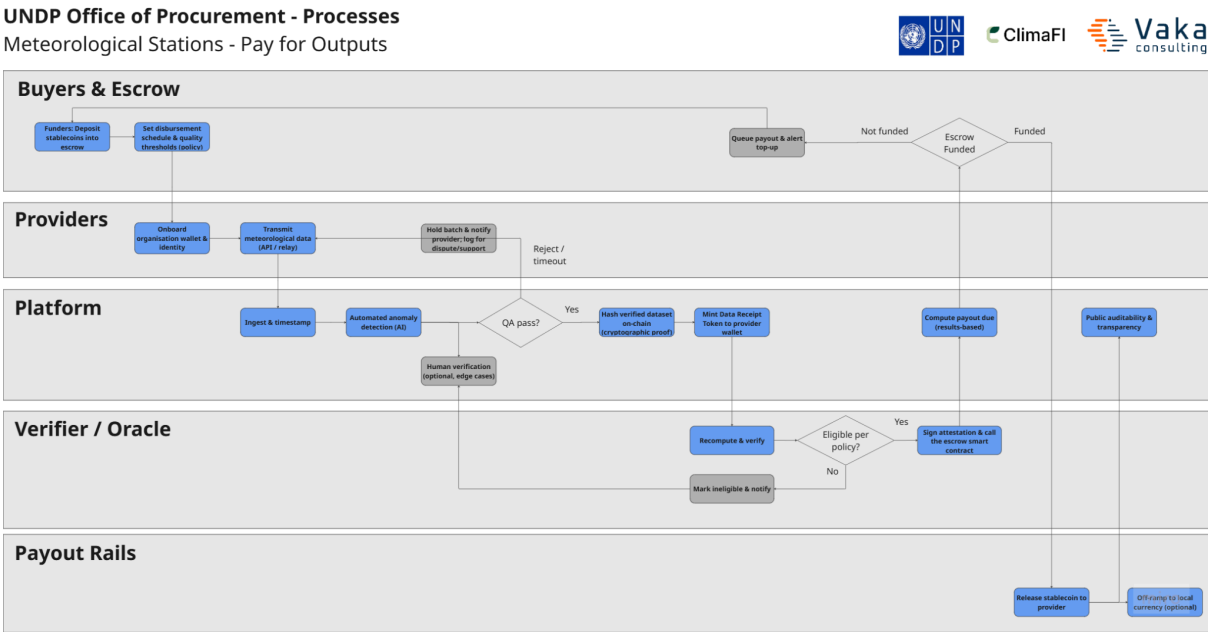


Fig 3: Simplified 5 Step Flow



Key Terms

- **Receipt Token:** A cryptographic token issued upon successful verification of meteorological data. Each token embeds a unique hash linking it immutably to the dataset, enabling transparent audits
- **Data Quality Verification:** The process of validating meteorological datasets through automated, rules-based verification (and optional human review) to confirm accuracy, completeness, and reliability against WMO standards
- **Verification Gate:** The logic checkpoint (Pass → Produce Token; Fail → No Token) that ensures donors only pay for observations that meet agreed quality standards.
- **Shadow Escrow:** An automated blockchain contract used in the pilot to simulate the holding and release of funds based on verified data, creating a "Digital Twin" of the financial flow
- **Stablecoin:** A digital currency pegged to a fiat currency (e.g., USD). *Note: In the pilot phase, stablecoins are simulated to test the payment rails without regulatory friction*

3.2 Core Features and Functionalities

Automated Data Quality Verification: A rules-based quality control engine that checks incoming station data against GBON/WIS2 metrics (timeliness, range, consistency) to flag anomalies without human intervention

Receipt Token Issuance: The minting of a unique "Data Receipt Token" on the blockchain for every valid dataset. This token embeds a cryptographic hash to create an immutable audit trail of performance, serving as a "Proof of Work" for the climate sector

Shadow Payment & Valuation Logic: A smart contract mechanism that calculates the *hypothetical* payout value for each valid receipt token. It displays "pending earnings" on a dashboard to demonstrate the viability of results-based finance (RBF) without transferring actual funds during the pilot

Auditability & Transparency: Publicly verifiable proofs linking specific weather data packets to their corresponding receipt tokens, ensuring donors can trust the data accounting system prior to depositing real capital

Stakeholder Dashboards: Role-based user interfaces:

- **NMHS View:** To monitor station uptime, data status, and accrued "Shadow Value"
- **Donor View (SOFF):** To view global compliance rates, verify data provenance, and estimate future funding requirements

3.3 Tech Stack and Tools

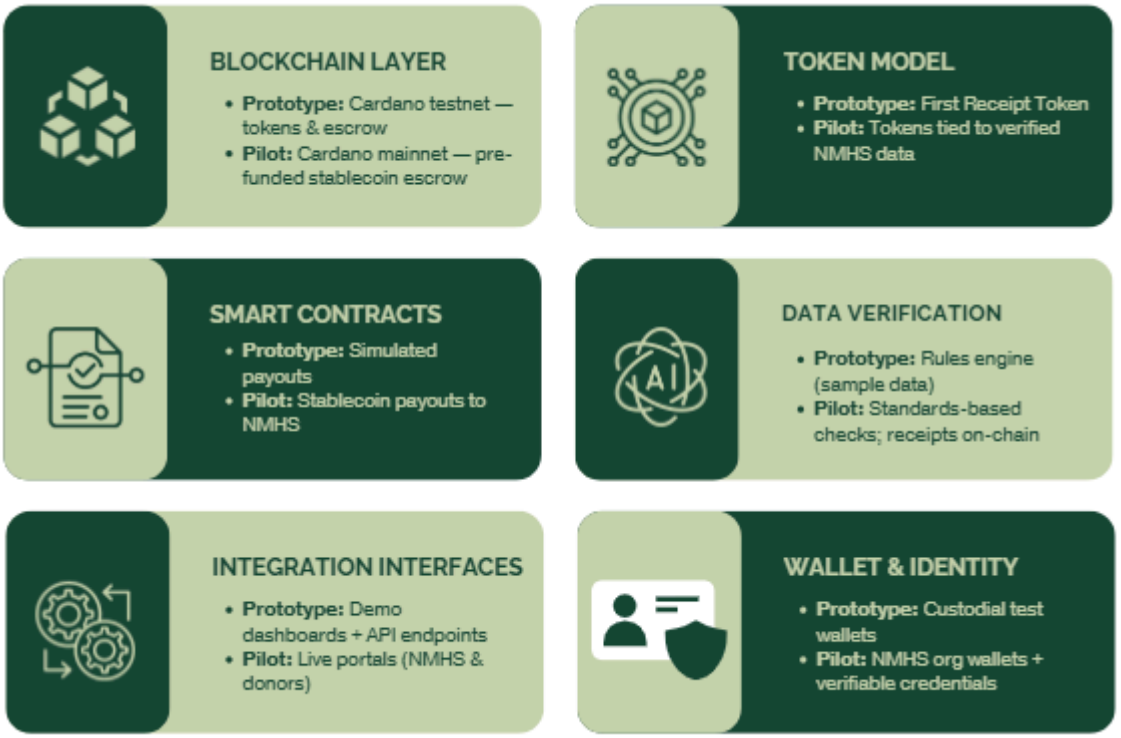


Fig 5: Tech Stack

- **Blockchain Layer**: Cardano (native tokens, smart contracts, on-chain hashing of reports)
- **Token Model:** Fungible or semi-fungible receipt tokens representing validated datasets. Metadata adheres to CIP-68 (Cardano Improvement Proposal) standards to ensure interoperability and efficient storage of meteorological attributes
- **Smart Contracts:** Written in Plutus. These contracts handle the "Shadow Escrow" logic, calculating theoretical payouts based on verification triggers
- **Data Verification Layer:** A custom engine (Python/Node.js) where data transmitted from meteorological equipment undergoes automated, rules-based verification (deterministic range/consistency/timeliness checks aligned to GBON/WIS2) with optional human oversight
- **Integration Interfaces:** MQTT/REST APIs for ingesting data from meteorological stations; React-based web dashboards for ministries and donor portals

- **Wallet & Identity Layer:** Secure organizational wallets for the NMHS. The system utilizes Decentralized Identifiers (DIDs) to authenticate data origin, preventing "spoofing" of weather stations

3.4 Blockchain Functionality

What blockchain-specific interactions are included (e.g., token minting, smart contracts, on-chain verification)?
Token Minting: The creation of "Receipt Tokens" on the Cardano blockchain. A strict Minting Policy ensures tokens can only be generated by the authorized Verification Engine when data passes quality checks.

On-Chain Hashing: Embedding a cryptographic hash of the off-chain data payload (e.g., the specific CSV/BUFR file) and a timestamp directly into the token metadata. This ensures data immutability and provenance.

Smart Contract Logic: Execution of "escrow-like" logic. The contract validates the presence of a Receipt Token in the wallet and updates the ledger state to reflect the "earned" value. In the pilot, this operates in "Shadow Mode" (simulated settlement) to prove the accounting model.

Identity Management: Use of organizational wallets to authenticate the NMHS as the authorized data publisher, ensuring that "Shadow Payments" are attributed only to verified government entities.

4. Weekly Workplan

Timeline (Week 1, 2, 3..)	**Focus / Category** (Blockchain Interaction, User Interface, Stakeholder Testing, SDG Integration Demo Readiness, etc.)	**Baseline Metric** (One meaningful blockchain function implemented (e.g., token minting, VC issuance, on-chain hash; etc.)
Week 1	**Challenge Deep Dive & Scope Finalisation** ● Align with the UNDP Office of Procurement on pilot context (target ministry, data types, reporting cadence). ● Map existing meteorological infrastructure & sample data flows. ● Agree on quality criteria (frequency, completeness, accuracy thresholds).	Signed-off requirements doc; example dataset collected for testing.
Week 2	**System Architecture & Standards Alignment** ● Draft system architecture (data pipeline, verification, tokenization, payment contracts). ● Confirm integration standards. ● Establish testnet environment.	Technical architecture diagram & testnet environment live.

Week 3	**Data Quality Verification Prototype** ● Develop automated checks model using sample historical data and standards ● Build a simple dashboard showing flagged anomalies. ● Define workflow for optional human review of flagged data.	Anomaly detection prototype functioning on sample data set.
Week 4	**Data Pipeline & API Integration** ● Connect simulated meteorological station feeds to the verification system. ● Standardize incoming data format & storage. ● Begin building an audit log of verification decisions.	End-to-end flow: data ingested → processed → flagged/approved.
Week 5	**Receipt Token Design & Issuance** ● Design token schema (metadata includes cryptographic hash, timestamp, provider ID). ● Implement code for token minting on testnet. ● Demonstrate first mint of a “Data Receipt Token” linked to verified dataset.	First testnet token minted with dataset hash embedded.
Week 6	**Results-Based Payment Logic** ● Design stablecoin escrow contract (funds locked by donor). ● Implement payout logic triggered by valid receipt tokens. ● Define payout schedule (e.g., weekly, monthly).	Testnet contract executes payout in stablecoin to provider wallet.
Week 7	**Integration of Token + Payment Contracts** ● Connect receipt tokens to payment contract as proof of verified data. ● Test multiple cycles of data → token → payout. ● Document edge cases (failed verification, duplicate data).	Demonstrated linkage between token issuance and payout logic.
Week 8	**Provider & Donor Dashboards** ● Develop a dashboard for providers (view tokens earned, balances). ● Develop donor dashboard (funds deposited, tokens verified, payments released). ● Connect wallets to dashboards.	First version of dashboards live and connected to testnet contracts.
Week 9	**Identity & Access Management** ● Integrate organizational wallet setup (PIN or credential system).	Provider identity linked to wallet on testnet.

	• Test provider onboarding flow (wallet creation, linking to ministry identity). • Explore verifiable credentials for provider identity.	
Week 10	**Stakeholder Testing & Feedback** • Conduct live demo for UNDP Office of Procurement and selected ministry (if applicable). • Capture feedback on usability, workflows, and reporting. • Adjust contracts/dashboards based on input.	Stakeholder demo completed; feedback incorporated.
Week 11	**Iteration & Refinement** • Improve UI/UX based on testing feedback. • Stress test smart contract logic with larger simulated data volumes. • Document operational manual for providers & donors.	Stable MVP workflow completed with improved UI & documentation.
Week 12	**MVP Finalization & Pilot Readiness** • Package final MVP for deployment in real-world pilot (data → token → payment). • Finalize monitoring & evaluation metrics for pilot. • Deliver presentation/demo for Accelerator closing session.	Fully functioning MVP with pilot-ready documentation and live demo.

5. Final Deliverables

5.1 Solution Demo (PoC)

A working or visual prototype that demonstrates the core idea. (To be scheduled and presented virtually to CO team and UNDP/Partner FPs in a weekly coordination meeting)

See Recording: *https://youtu.be/itMzyzZMZL8*

5.2 Testing & Feedback Documentation

Summary of user testing activities, challenges, and feedback.

Agile Co-Creation Process Testing was not conducted in isolation but through a structured "Co-Creation Cycle" involving weekly synchronization with our core partners to align on needs, timelines, and institutional constraints.

- **Weekly Technical Sprints (UNDP Office of Procurement, SOFF & Solution Makers):** Focused on the realities of data ingestion.
- **Strategic Alignment (UNDP Tanzania Country Office):** Focused on regulatory and political alignment.

- **Compliance Reviews (SOFF & WMO Standards):** Focused on the "Oversight" requirements.

5.3 Feedback Integration Summary

Description of what changes were made based on testing and feedback.

1. Pivot to "Digital Audit" (SOFF Feedback)

- **Feedback:** SOFF stakeholders emphasized that while blockchain is a promising rail for future payments, the immediate gap is a lack of trusted data accounting.
- **Integration:** We shifted the solution's primary function from "Payment Automation" to "Digital Auditability." The system now prioritizes generating immutable "Data Receipt Tokens" to prove GBON compliance first, treating the financial automation as a secondary "Shadow" layer. This ensures the technology solves the immediate trust problem before attempting to solve the money problem.

2. Asynchronous Verification Logic (Technical Reality)

- **Feedback:** Testing with TMA engineers revealed that strict "Real-Time" requirements would unfairly penalize rural stations that often face connectivity lags.
- **Integration:** We re-engineered the verification engine to support "Asynchronous Validation." The system now accepts valid data packets up to 24 hours late without penalty, ensuring that stations with intermittent internet are still rewarded for their data as soon as connection is restored.

3. "Shadow Mode" for De-Risking (Regulatory Feedback)

- **Feedback:** The Country Office highlighted the significant legal complexity of moving live crypto-assets through government accounts without prior Central Bank frameworks.
- Integration: We adopted a "Shadow Mode" Strategy. Instead of forcing live payments immediately, the pilot will use a simulated ledger to calculate earnings. This allows us to "start with data" to build institutional trust and generate the empirical evidence required to apply for the Regulatory Sandbox in Phase 2.

5.4 Pilot-Ready Solution

A stable and functional version of the solution, ready for pilot deployment.

We have successfully developed a prototype, and ready to proceed to pilot pending funding.

5.5 Confirmed Pilot Partnership(s)

The CO of Tanzania, confirms participation in the pilot.

We've had significant engagement with SOFF and WMO throughout this accelerator.

1. **UNDP Tanzania Country Office (Challenge Owner)**
- Status: Confirmed

- Role: The Country Office has confirmed participation for the pilot of this solution.

2. **Systematic Observations Financing Facility (SOFF) (Strategic Partner)**

- **Status:** High-Level Engagement
- **Role:** We have maintained significant engagement with SOFF and WMO throughout the accelerator. They serve as the "Standard Setter," providing the GBON compliance metrics that our verification engine is built upon. Their role in the pilot is to validate the "Oversight Dashboard" as a potential global standard for results-based monitoring.

5.6 Stakeholder Alignment Matrix

Defined roles, responsibilities, and coordination plan with key partners.

Stakeholder / Partner	Role in the Pilot	Key Responsibilities	Level of Involvement	Engagement Modality	Focal Point / Contact Person
NMHS (National Met Services)	**Primary User / Provider**	transmitting station data; reviewing verification logs; testing dashboards; providing user feedback[1111] 1111	**Lead**	Pilot Agreement / MOU	National Director (TBD)
UNDP Country Office	**Challenge Owner**	Coordinating local government alignment; facilitating access to ministries; aligning pilot with national	**Lead**	Internal Project	CO Rep (TBD)

		priorities[1212] [1212].			
ClimaFI	**Solution Maker (Verification)**	Developing the off-chain verification engine; building user dashboards; implementing anomaly detection	**Lead**	Partnership Agreement	Matt
Vaka Consulting	**Solution Maker (Blockchain)**	Developing smart contracts; deploying Cardano rail; managing wallet/identity integration	**Lead**	Partnership Agreement	Phil Lewis
SOFF (Systematic Observations Financing Facility)	**Observer / Future Funder**	Providing requirements for GBON compliance; reviewing pilot outcomes for future scaling potential.	**Advisory**	Strategic Partnership	SOFF Rep (TBD)

WMO (World Met Org)	**Standards Body**	Advising on WIS 2.0 / GBON data standards; ensuring technical alignment with global zystems[16161616]	**Advisory**	Advisory Support	WMO Rep (TBD)

5.7 Budget Proposal (Pilot)

Provide a budget breakdown for pilot implementation. Include estimated cost, unit, quantity, and funding source.

Budget Line Item	Unit / Description	Quantity	Unit Cost (USD)	Total Cost (USD)	Funding Source
1. Personnel & Technical Development					
Blockchain Architect (Lead)	Days (Smart Contract logic, Token Registry, Shadow Escrow)	60	$600	**$36,000**	Pending (Grant)
Data Scientist (Verification)	Days (ClimaFI Algorithm tuning, Anomaly detection, Historical data testing)	40	$550	**$22,000**	Pending (Grant)

Full Stack Developer (Frontend)	Days (TMA Provider Dashboard & SOFF Oversight Dashboard)	30	$500	**$15,000**	Pending (Grant)
Project Manager	Months (Stakeholder liaison, Agile management, Reporting)	6	$3,000	**$18,000**	Pending (Grant)
TMA Engineering Support	Days (Station integration support & server access)	20	$250	**$5,000**	**Confirmed (In-Kind)**
2. Operations & Infrastructure					
Legal & Regulatory Counsel	Flat Fee (Sandbox Application & Data Sovereignty Opinion)	1	$8,000	**$8,000**	Pending (Grant)
Cloud Infrastructure	Months (AWS/Google Cloud, API Gateways, Node Hosting)	6	$1,500	**$9,000**	Pending (Grant)

Security Audit	Flat Fee (Smart Contract & API Security Review)	1	$4,000	**$4,000**	Pending (Grant)
3. Capacity Building & Travel					
Stakeholder Workshops	Events (Kick-off & Final Demo Day in Dar es Salaam)	2	$3,000	**$6,000**	Pending (Grant)
Travel	Trips (Site visits to station hubs for calibration)	4	$1,000	**$4,000**	Pending (Grant)
4. Contingency					
Miscellaneous / FX Buffer	% of Total Budget (For currency fluctuation & unforeseen costs)	1	$3,000	**$3,000**	Pending (Grant)
GRAND TOTAL				**$130,000**	

5.8 Impact Measurement Framework

5.8.1 Contribution to relevant SDGs and indicators

Goal	Target	Specific Indicator	Project Contribution Metric
SDG 13: Climate Action	**13.1** Strengthen resilience and adaptive capacity.	**13.1.2:** Number of countries that adopt and implement national disaster risk reduction strategies.	**Metric:** Number of TMA weather stations (27) upgraded to provide *verified, real-time data* for the National Early Warning System.
SDG 9: Industry, Innovation & Infrastructure	**9.1** Develop quality, reliable, sustainable and resilient infrastructure.	**9.1.1:** Proportion of the rural population who live within 2 km of an all-season road (Adapted to **Information Access**).	**Metric:** % of Pilot Stations in rural/remote areas successfully transmitting verifiable data to the central dashboard without manual intervention.
SDG 17: Partnerships	**17.6** Enhance North-South and South-South cooperation on technology.	**17.6.1:** Number of science and/or technology cooperation agreements and programmes.	**Metric:** Execution of one (1) "Shadow Payment" Data Sharing Protocol signed between TMA (Provider) and UNDP/SOFF (Funder).

5.8.2 Key Performance Indicators (KPIs)

KPI	***Measurement Method***	***Target (Pilot Phase)***	***Notes***
Station Connectivity	*Number of stations successfully ingesting data into the Verification Node.*	***27 Stations***	*The primary goal is simply to get the data flowing and visible.*
Verification Latency	*Time elapsed between Data Arrival and Token Generation.*	***< 60 Minutes***	*A realistic target for rural infrastructure (allows for batch processing).*
Ledger Consistency	*Consistency between TMA's internal data logs and the Blockchain "Shadow Wallet".*	***100% Match***	*The critical success factor is proving that the accounting is accurate.*
Reporting Frequency	*Generation of "Shadow Settlement Reports" for stakeholders.*	***Monthly***	*Demonstrates the administrative capacity to manage a Pay-for-Results cycle.*

5.8.3 Monitoring & Reporting Plan (Pilot)

a) Frequency:

- *Bi-Weekly technical sprint reviews between Solution Makers and UNDP*
- *Monthly stakeholder progress reports for the Ministry (NHMS) and Donors (simulated).*
- *Real-time data flow monitoring via the project dashboards.*

b) Responsible Parties:

- *UNDP Country Office: Overall project coordination and stakeholder liaison*
- *Solution Makers (ClimaFI / Vaka): Technical performance monitoring (uptime, verification rates) and bug reporting*
- *NHMS: Monitoring of station data transmission and dashboard usability feedback*

c) Format:

- ***Live Dashboards:*** *Visualizing data ingestion, verification pass/fail rates, and "shadow" earnings*
- ***Sprint Reports:*** *PDF/Slide summaries of features built, bugs fixed, and upcoming priorities*
- ***Feedback Logs:*** *Shared document tracking user feedback from NHMS staff and resolution status*

d) Tools:

- ***Project Management:*** *GitHub/Jira (for technical tasks)*
- ***Visualization:*** *ClimaFI/Vaka Custom Dashboards (for data & token metrics)*
- ***Communication:*** *Google Meet for coordination meetings; Shared Drive for documentation*

5.8.4 Risks Assessment & Mitigation Plan

Risks	*Mitigation*
Regulatory hurdles for crypto payments: Difficulty in legally transferring stablecoins to government wallets	Shadow Payment Model: Operate a "shadow" accounting system during the pilot to prove value without regulatory friction. Engage licensed on/off-ramp partners for the post-pilot phase
Data Quality Issues: Station data failing to meet GBON standards (gaps, noise) resulting in zero "earnings"	Feedback Loop: The dashboard will clearly visualize why data failed (e.g., "missing timestamps"), helping NHMS staff fix equipment issues.
Connectivity & Power: Remote stations going offline, breaking the data stream	Asynchronous Verification: The system accepts delayed data (within defined windows) to account for connectivity intermittency

5.9 Post-Acceleration Roadmap & Sustainability Plan

5.9.1 Phased Implementation Timeline / Milestones

Short Term (0–6 months)	Medium Term (6–18 months)	Long Term (18+ months)
Deploy "Shadow Escrow" smart contracts on Cardano mainnet (simulated settlement). **Validate Data Stream** from the 27 target TMA stations	**Admission to Regulatory Sandbox** (Bank of Tanzania) for live stablecoin testing. **First "Live" Micropayment:** Execute a controlled	**Global Scale-Up:** Integration as the official "Compliance Layer" for the SOFF Global Portfolio. **Cross-Border Replication:** Deploy the validated stack to

against GBON/WIS2 standards. **Publish "Proof of Value":** Generate monthly "Shadow Settlement Reports" for the Ministry of Finance.	settlement to a ring-fenced wallet. **Commercial Onboarding:** Secure first commercial data purchase (e.g., Index Insurance) to test revenue diversity.	2-3 additional LDCs (e.g., Rwanda, Malawi). **Financial Sustainability:** Activate the Protocol Fee on all transaction volumes to fund system maintenance.

Pilot/Proof of Concept (PoC) Implementation Plan

Project Title: Results-Based Payments for Meteorological Data (RBP-MetData)

Location: Tanzania (27 Surface Stations, 5 Upper-Air Stations)

Duration: 6 Months (Shadow Execution Phase)

Budget: $130,000 USD

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1. Strategic Objectives

The primary goal is to validate the **"Digital Audit"** capability of the system without exposing stakeholders to regulatory or financial risk.

- **De-Risking:** Prove that the verification engine works on real data (handling latency/noise) before live funds are attached.
- **Shadow Accounting:** Demonstrate to the Ministry of Finance and Donors (SOFF) exactly how much revenue *would* have been earned based on the data quality provided.
- **Regulatory Evidence:** Generate the empirical performance data required to apply for the **Bank of Tanzania Fintech Regulatory Sandbox** for Phase 2 (Live Payments).

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2. Governance & Partnerships

Partner	Role	Responsibilities
UNDP Tanzania	**Challenge Owner**	Political alignment, Sandbox application coordination, Ministry liaison.

TMA (NMHS)	**Problem Owner**	Providing data access (API/FTP), engineering support, and user feedback.
SOFF Secretariat	**Strategic Partner**	Defining GBON compliance metrics; primary user of the "Oversight Dashboard."
ClimaFI / Vaka	**Solution Maker**	Technology deployment, smart contract management, and system maintenance.

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3. Phased Implementation Roadmap

This roadmap covers the **6-Month Execution Phase** (post-accelerator).

Phase 1: Stabilization & Baseline (Month 1)

- **Focus:** Ensuring the "Plumbing" works.
- **Activities:**
  - Deploy the Verification Node to the production environment.
  - Establish stable API/MQTT connections with all 27 target stations.
  - **Baseline Metric:** Monitor data availability for 30 days to establish a "Quality Baseline" (e.g., current uptime is 60%; target is >90%).
- **Deliverable:** System Stability Report.

Phase 2: The "Shadow Payment" Cycle (Months 2–3)

- **Focus:** Proving the "Accounting" works.
- **Activities:**
  - Activate the **Shadow Escrow Smart Contract**.
  - **Minting:** Every valid data packet triggers the minting of a "Data Receipt Token" on the Cardano blockchain.
  - **Valuation:** The dashboard calculates the theoretical dollar value of these tokens based on the agreed rate card.
- **Deliverable:** Monthly "Shadow Settlement Reports" presented to TMA and Ministry of Finance.

Phase 3: Stress Testing & Optimization (Month 4)

- **Focus:** Proving "Resilience."
- **Activities:**
  - **Adversarial Testing:** Simulate network outages, duplicate data packets, and sensor malfunctions to ensure the Verification Engine rejects bad data (Zero False Positives).

    - **Asynchronous Test:** Verify that data arriving 24 hours late (due to rural connectivity) is correctly verified and paid retroactively.
- **Deliverable:** Security & Resilience Audit.

Phase 4: Regulatory Sandbox & Evaluation (Months 5–6)

- **Focus:** Preparing for "Live Money."
- **Activities:**
    - Compile the 6-month evidence dossier (Transaction Logs, Uptime Improvements, Payment Accuracy).
    - Submit the application to the **Bank of Tanzania Fintech Regulatory Sandbox**.
    - Host the **Final Stakeholder Demo** with SOFF to secure endorsement for Phase 2 funding.
- **Deliverable:** Final Impact Report & Signed Phase 2 MoU.

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4. Technical Scope (The "Shadow Stack")

- **Ingestion:** Python/Node.js gateway accepting BUFR/CSV formats from TMA servers.
- **Verification:** Rules-based engine checking **Timeliness** (<60 min), **Consistency** (Internal/Spatial), and **Completeness** (GBON Variables).
- **Blockchain: Cardano Mainnet** (using metadata for audit trails) but with **Testnet Stablecoins** (or Shadow Accounting tokens) for settlement simulation.
- **Dashboards:**
    - *TMA View:* Station Health, Data Upload Logs, Shadow Earnings Wallet.
    - *SOFF View:* Global Map, Compliance %, Funding Efficiency Metrics.

5.9.2 Budget for Phased Implementation (post-acceleration)

Budget Line Item	Unit / Description	Quantity	Unit Cost (USD)	Total Cost (USD)	Funding Source
1. Personnel & Technical Development					
Blockchain Architect (Lead)	Days (Smart Contract logic, Token Registry, Shadow Escrow)	60	$600	**$36,000**	Pending (Grant)

Data Scientist (Verification)	Days (ClimaFI Algorithm tuning, Anomaly detection, Historical data testing)	40	$550	**$22,000**	Pending (Grant)
Full Stack Developer (Frontend)	Days (TMA Provider Dashboard & SOFF Oversight Dashboard)	30	$500	**$15,000**	Pending (Grant)
Project Manager	Months (Stakeholder liaison, Agile management, Reporting)	6	$3,000	**$18,000**	Pending (Grant)
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Cloud Infrastructure	Months (AWS/Google Cloud, API Gateways, Node Hosting)	6	$1,500	**$9,000**	Pending (Grant)
Security Audit	Flat Fee (Smart Contract & API Security Review)	1	$4,000	**$4,000**	Pending (Grant)
3. Capacity Building & Travel					
Stakeholder Workshops	Events (Kick-off & Final Demo Day in Dar es Salaam)	2	$3,000	**$6,000**	Pending (Grant)
Travel	Trips (Site visits to station hubs for calibration)	4	$1,000	**$4,000**	Pending (Grant)
4. Contingency					
Miscellaneous / FX Buffer	% of Total Budget (For currency fluctuation & unforeseen costs)	1	$3,000	**$3,000**	Pending (Grant)

GRAND TOTAL				**$130,000**	

5.9.3 Financial Sustainability Model

How will the solution generate income post-pilot? Pricing model (e.g. subscriptions, licensing, transaction fees) and when it kicks in.

Here are the revised sections. I have removed the hard numbers (dollar amounts, specific percentages, station counts) and softened the language to be **indicative of the strategy** without making binding assumptions about partner commitments or regulatory outcomes.

5.9.1 Phased Implementation Timeline / Milestones

Short Term (0–6 months)	Medium Term (6–18 months)	Long Term (18+ months)
Deploy "Shadow" Ledger on blockchain testnet to validate the accounting logic.	**Regulatory Engagement:** Initiate dialogue with financial regulators regarding future payment mechanisms.	**Global Scale-Up:** Position solution as a verified compliance tool for broader SOFF investment portfolios.
Data Validation: Verify live data streams from a representative sample of NMHS stations against GBON standards.	**Payment Trials:** Explore executing the first controlled, low-value financial settlement to a pilot wallet.	**Cross-Border Replication:** Adapt the technical stack for deployment in additional LDCs or SIDS.
Proof of Value: Generate "Shadow Settlement Reports" demonstrating potential earnings to key stakeholders.	**Commercial Exploration:** Test demand for verified data products with insurance or agricultural partners.	**Financial Sustainability:** Operationalize long-term revenue models (e.g., transaction fees) to fund system maintenance.

Pilot/Proof of Concept (PoC) Implementation Plan

Project Title: Results-Based Payments for Meteorological Data (RBP-MetData)

Location: Tanzania (Selected Pilot Stations)

Duration: 6 Months (Targeted Pilot Phase)

1. Strategic Objectives

The primary goal is to validate the technical feasibility of **"Digital Audit"** and **"Shadow Accounting"** mechanisms.

- **Technical Validation:** Prove that the verification engine can successfully ingest and validate real-world station data against WMO standards.
- **Shadow Accounting:** Demonstrate to stakeholders how a results-based payment model would function by calculating theoretical revenue based on data quality.
- **Evidence Gathering:** Collect performance data to support future dialogues regarding regulatory approval for automated payments.

2. Governance & Partnerships

- **UNDP Tanzania:** Strategic oversight and alignment with national digital priorities.
- **TMA (NMHS):** Technical collaboration for data access and user feedback on dashboards.
- **SOFF Secretariat:** Guidance on GBON compliance metrics and reporting standards.
- **Technology Partners:** Provision and maintenance of the verification and blockchain infrastructure.

3. Phased Implementation Roadmap

Phase 1: Connectivity & Baseline

- **Focus:** Establishing stable data feeds.
- **Activities:** Connect the verification node to selected station APIs and establish a baseline for data quality and transmission latency.

Phase 2: The "Shadow Payment" Cycle

- **Focus:** Proving the accounting logic.
- **Activities:** Activate the smart contracts in "Shadow Mode" to mint receipt tokens and calculate theoretical values. Present findings to stakeholders via the dashboard.

Phase 3: Resilience Testing

- **Focus:** System stability.
- **Activities:** Test the system’s ability to handle data gaps, latency, and connectivity interruptions without failing.

Phase 4: Evaluation & Reporting

- **Focus:** Assessment for scale.
- **Activities:** Compile a final report detailing system accuracy, cost-efficiency, and readiness for future financial integration.

5.9.3 Financial Sustainability Model

How will the solution generate income post-pilot?

The solution aims to transition from grant-funded pilot to a self-sustaining infrastructure through a potential hybrid revenue model.

1. Transaction Fee Model

- **Concept:** The platform may implement a small operational fee on future climate finance transactions routed through the smart contracts.
- **Logic:** As funding volumes scale, a percentage-based fee would support ongoing system maintenance, security updates, and infrastructure costs, ensuring the platform remains sustainable without perpetual grant dependence.

2. SaaS / Licensing & Premium Audits

- **Concept:** While core data remains accessible, premium features—such as deep-tier forensic audits, real-time anomaly detection, and automated ESG certification reports—could be offered via a licensing model.
- **Target Audience:** Major institutional donors, insurers, or development agencies requiring high-assurance data verification.

5.9.4 Partnership and Co-Financing Commitments

Which partners will be supporting the project (technical, financial, institutional) and what would be their contributions?

SOFF (Target): *Primary source of long-term "Pay-for-Results" capital*

UNDP: *Operational support.*

WMO: *Endorsement of the verification standards (GBON compliance)*

5.9.5 Fundraising Plan

Plan for securing funding through grants, ecosystem funds, investors or other sources. Targeted sources and fundraising timeline.

Plan for securing funding through grants, ecosystem funds, and investors.

Phase 1: Pilot & Evidence (Short Term)

- **Source:** Innovation Grants / Accelerator Funding.
- **Purpose:** To fund the technical deployment of the "Shadow Mode" pilot and initial stakeholder workshops.

Phase 2: Operational Scale-Up (Medium Term)

- **Source:** Climate Finance Facilities / Multi-Partner Trust Funds.
- **Purpose:** To expand the network coverage, enhance verification logic, and support the legal/regulatory work required for live payments.

Phase 3: Ecosystem Expansion (Long Term)

- **Source:** Blockchain Ecosystem Funds / Web3 Climate Grants.
- **Purpose:** To open-source the verified protocol and support developer communities in replicating the model across other regions.

5.11 Legal & Compliance Considerations

Summary of key legal, regulatory, or compliance considerations.

It doesn't break any legal constraint from the country or WMO… Follows the Gbon compliance and Tanzania is a member state.

Operational/Coordination (Timing): *The new "Universal Data Collection System" for Tanzania is currently in procurement. The pilot must decide whether to wait for this system or implement a "data proofing" layer alongside the existing infrastructure.*

Financial (Public Financial Management): *There are legal/policy restrictions on direct payments to government agencies. TMA funds usually go through the Ministry of Finance. Stakeholders have suggested routing funds through UNDP Country Offices to ensure they are used for station maintenance rather than general budget absorption.*

Technical (Data Integration): *Determining where the blockchain "sits" in the data sharing architecture. The data shall be verified before it reaches WMO and to be useful for SOFF verification. The data structure will take into account WMO systems (WIS 2.0) which use the protocols (MQTT/BUFR) and will be compatible.*

Sustainability (Consumables): *High operational costs (e.g., $150/launch for upper-air stations) mean that if payment flows are delayed or insufficient, stations will go offline, causing a failure in GBON compliance and stopping future payments.*

5.12 Visibility Assets / Communication Materials

1–2 slides, visuals, or storylines for use by UNDP or partners to promote your solution.

5.13 Pitch Deck

Concise and compelling presentation for stakeholders, funders, and partners. (Please upload to your google drive folder)

https://docs.google.com/presentation/d/16vpDn_Y7gL2K77kf1XYFyzof3VtB0bKT/edit?usp=drive_link&ouid=111398004742958468307&rtpof=true&sd=true

https://docs.google.com/document/d/13y0krTtHNH_hPYIKJBDhza0Aw4Jtxngj/edit?usp=drive_link&ouid=111398004742958468307&rtpof=true&sd=true

5.14 Final Demo Presentation

An engaging walkthrough of the live solution and its implementation roadmap at the end of the program.

https://youtu.be/itMzyzZMZL8 (sample)

Deliverable Completion Table

Please indicate your intended completion week (or date) for each deliverable. You can revise this plan as the project evolves.

Deliverable	Target Completion (Week # or Date)
1. Challenge Definition (incl. local context, SDG contributions, partners, impact)	+ 1
2. User Mapping & Testing Strategy (personas, user stories, feedback loop)	+ 3
3. Solution Overview (core features, tech stack, blockchain functionality)	+ 10
5.1 Solution Demo (PoC)	+ 5
5.2 Testing & Feedback Documentation	+ 8
5.3 Feedback Integration Summary	+ 2
5.4 Pilot-Ready Solution	+ 4
5.5 Confirmed Pilot Partnership(s)	+ 4
5.6 Stakeholder Alignment Matrix	+ 2
5.7 Budget Proposal (Pilot)	+ 8
5.8 Impact Measurement Framework (KPIs, SDG indicators, risk plan)	+ 2
5.9 Sustainability Plan & Post-Acceleration Roadmap (implementation timeline, budget, partners, fundraising)	+ 2
5.11 Legal & Compliance Mapping	+2
5.12 Visibility Assets / Communication Materials	+2
5.13 Pitch Deck	+1